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S. No.	Practical	Date	Sign
1	Write a menu-driven calculator program using functions		
2	Calculate the factorial of a number using functions.		
3	Find fibonacci series upto n number entered by the user using functions.		
4	Write and execute all built-in functions. Like: int(), str(), float(), input(), eval(), max(), abs(), type(), len(), round(), range(), bytes(), chr(), dir(), exec(), format, hex(), ascii, etc. (Write small description also for each function)		
5	Write a program to display square root of a number using math module.		
6	Write a simple calculator program which returns multiple statements.		
7	Write three python programs separately which shows the concept of Default arguments, Keyword arguments and Variable Length Argument.		
8	Write a program and pass a list to a function that calculate the arithmetic mean/average of list elements entered by the user at run time.		
9	Write a program that takes the string entered by the user and count its vowels. Passing string to a function concept.		
10	Write a program in python to calculate factorial of a number and generate fibonacci series using the concept of recursion		
11	Write a program to count the number of a particular character from the file		
12	Write a program to find no. of times a particular word is available in the file or not.		
13	Write a program to read a text file line by line and display each word separated by #.		
14	Write a program to read a text file and display the number of vowels/consonants/uppercase/lowercase characters in the file.		
15	Write a program to remove all the lines that contain the character 'a' in a file and write it to another file.		
16	WAP to create a binary file with name and roll number. Search for a given roll number and display the name, if not found display appropriate message.		
17	WAP to create a binary file with roll number, name and marks. Input a roll number and update the marks		

18	Write a program to create a CSV file as “student.csv” for maintaining a student record which includes Student Name, Roll No., Gender and Marks.		
19	Write a program to read and display the contents of “student.csv” file.		
20	Write a program to create a CSV file by entering user id and password, read and search the password for given user id.		
21	Write a program for random number generator that generates random numbers between 1 and 6(simulates a dice).		
22	Write a python program to implement a stack using list. (Menu driven program for PUSH, POP, Display and EXIT options)		

23	Write a python program to create a stack of EVEN no. by pushing only even numbers from the list of integers entered by the user.		
24	Write a python program to create a stack, by pushing only vowels from the string entered by the user.		
25	Write a python program to implement a Stack for the Car details (Car_no, Car_Make, Price). That is now each item node of the stack contains two types of information: a Car No. , its make, and the price of the car. Just implement menu driven PUSH, POP and DISPLAY operations.		
26	- Create a stack to manage student records. Each student record is represented as a dictionary containing attributes like student ID, name, and GPA. - Implement a `push_student` method to push student records onto the stack that GPA above 60. - Implement a `pop_student` method to pop the top student record from the stack.		
27	Create a student table and insert 10 row data. Implement the following SQL commands on the student table: - o ALTER table to add new attributes / modify data type / drop attribute - o UPDATE table to modify data - o ORDER By to display data in ascending / descending order - o DELETE to remove tuple(s) - o GROUP BY and find the min, max, sum, count and average		
28	Consider the following tables Product and Client. Write SQL commands for the statements (i) to (iii) and give outputs for SQL queries (i) to (vi). (i) To display the details of those Clients whose city is Delhi. (ii) To display the details of Products whose Price is in the range of 50 to 100 (both values included). (iii) To display the details of those products whose name ends with ‘Wash’ (iv) select distinct City from CLIENT; (v) select Manufacturer, max(Price), min(Price), count(*) from PRODUCT group by Manufacturer; (vi) select Product Name, Price * 4 from PRODUCT;		

29	(a) Display the names of the students who have grade ‘C’ in either Game1 or Game2 or both. (b) Display the number of students getting grade ‘A’ in Cricket. (c) Display the names of the students who have the same game for both Game1 and Game2. (d) Display the game taken up by the students, whose name starts with ‘A’. (e) Add a new column named ‘Marks’. (f) Assign a value 200 for marks for all those who are getting grade ‘B’ or grade ‘A’ in both Game1 and Game2.																																																											
30	Consider the following tables: COMPANY and MODEL: Comp_ID is the Foreign Key referencing Comp_ID of the Company table. Write SQL commands for queries (i) to (iv) and output for (v) and (vi). (i) To display details of all models in the Model table in ascending order of DateOfManufacture. (ii) To display details of those models manufactured in 2011 and whose Cost is below 2000. (iii) To display the Model_ID, Comp_ID, Cost from the table Model, CompName and ContactPerson from Company table, with their corresponding Comp_ID. (iv) To decrease the cost of all the models in the Model table by 15%. (v) select count(distinct CompHO) from Company; (vi) select CompName, contact(‘Mr.’,ContactPerson) from Company where CompName ends with ‘a’;																																																											
31	Write output for (i) to (iii) based on the tables ‘Watches’ and ‘Sale’ given below. Table: Watches <table><tr><th>Watchid</th><th>Watch_Name</th><th>Price</th><th>Type</th><th>Qty_Store</th></tr><tr><td>W001</td><td>HighTime</td><td>10000</td><td>Unisex</td><td>100</td></tr><tr><td>W002</td><td>LifeTime</td><td>15000</td><td>Ladies</td><td>150</td></tr><tr><td>W003</td><td>Wave</td><td>20000</td><td>Gents</td><td>200</td></tr><tr><td>W004</td><td>HighFashion</td><td>7000</td><td>Unisex</td><td>250</td></tr><tr><td>W005</td><td>GoldenTime</td><td>25000</td><td>Gents</td><td>100</td></tr></table> Table: Sale <table><tr><th>Watchid</th><th>Qty_Sold</th><th>Quarter</th></tr><tr><td>W001</td><td>10</td><td>1</td></tr><tr><td>W003</td><td>5</td><td>1</td></tr><tr><td>W002</td><td>20</td><td>2</td></tr><tr><td>W003</td><td>10</td><td>2</td></tr><tr><td>W001</td><td>15</td><td>3</td></tr><tr><td>W002</td><td>20</td><td>3</td></tr><tr><td>W005</td><td>10</td><td>3</td></tr><tr><td>W003</td><td>15</td><td>4</td></tr></table> i. select quarter, sum(qty_sold) from sale group by quarter; ii. select watch_name,price,type from watches w, sale s where w.watchid!=s.watchid; iii. select watch_name, qty_store, sum(qty_sold), qty_store-sum(qty_sold) “Stock” from Watches w, sale s where w.watchid=s.watchid group by s.watchid;	Watchid	Watch_Name	Price	Type	Qty_Store	W001	HighTime	10000	Unisex	100	W002	LifeTime	15000	Ladies	150	W003	Wave	20000	Gents	200	W004	HighFashion	7000	Unisex	250	W005	GoldenTime	25000	Gents	100	Watchid	Qty_Sold	Quarter	W001	10	1	W003	5	1	W002	20	2	W003	10	2	W001	15	3	W002	20	3	W005	10	3	W003	15	4		
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32	Use MySQL-Python Connectivity and Insert record in Employee Table entered by the user and display the inserted record in output windows. Attributes of the employee table: EmpID, EmpName, EmpDept, EmpDesig, DOJ(date of joining), Salary																																																											
33	Use MySQL-Python Connectivity and Insert 5 record in Employee Table by using executemany() function. Attributes of the employee table: EmpID, EmpName, EmpDept, EmpDesig, DOJ(date of joining), Salary																																																											
34	Write a python program to insert 5 records in Employee table. Take these 5 records as an input from the user (One record at a time). Note																																																											

	the following to establish connectivity between Python and MySQL: Username is root Password is 12345 The table exists in a MySQL database named company. The table has five attributes (Emp_ID, Emp_Name, DOJ, Gender, Salary)		
35	Write a python program to update name of employee in Employee table whose employee id is 'E1001' (Take updates name as an input from the user). Note the following to establish connectivity between Python and MySQL: Username is root Password is 12345 The table exists in a MySQL database named company. The table has five attributes (Emp_ID, Emp_Name, DOJ, Gender, Salary)		